

Algorithm Brainstorming				
Criteria	Weight	Insertion Sort	Cycle Sort	Gnome Sort
Conceptually different	5	0	+	0
Easy to code from scratch	5	0	-	0
Easy to understand and explain	4	0	0	0
Suitability for non-numeric data	4	0	0	0
Length of code	2	0	-	0
Sum of "+" (Weight)(+)		0	1x5 = 5	0
Sum of "0" (Weight)(0)		5x0 = 0	1x0 = 0	5x0 = 0
Sum of "-" (Weight)(-)		0	1x(-5) + 1x(-2) = -7	0
Net (out of 20)		0	5 + 0 + (-7) = -2	0

<https://www.geeksforgeeks.org/insertion-sort-algorithm/>

<https://www.geeksforgeeks.org/dsa/cycle-sort/>

<https://www.geeksforgeeks.org/dsa/gnome-sort-a-stupid-one/>

Algorithm Selection				
Criteria	Weight	Insertion Sort	Cycle Sort	Gnome Sort
Stable	4	0	-	0
In-place	5	0	0	0
Minimises number of swaps	5	0	+	-
Suitability for large datasets	4	0	0	0
Average and worst case Time Complexity	2	0	0	0
Sum of "+" (Weight)(+)		0	1x5 = 5	0
Sum of "0" (Weight)(0)		5x0 = 0	3x0 = 0	4x0 = 0
Sum of "-" (Weight)(-)		0	1x(-4) = -4	1x(-5) = -5
Net (out of 20)		0	5 + 0 + (-4) = 1	0 + 0 + (-5) = -5

User Interface

Sign in

Register

Exercise 1

Exercise 2

Exercise 3

Product Spreadsheet

Welcome! Add products by keying in the information and then click add

Enter ID:

Enter Name:

Enter Price:

Add

Sort by ID

Sort by Name

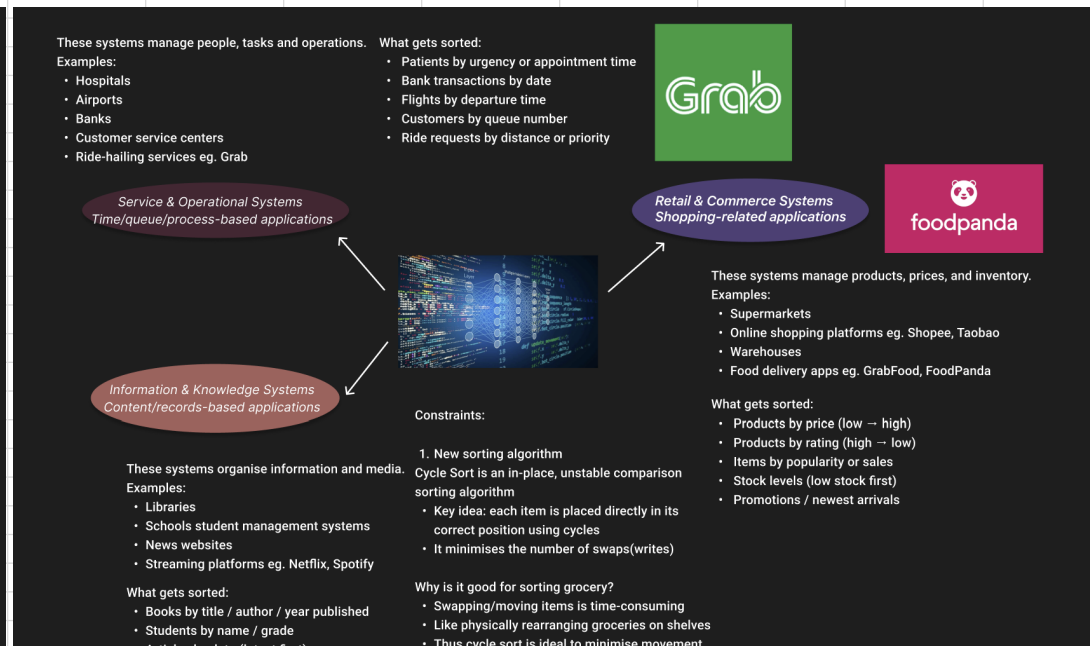
Sort by Price

Clear

ID	Name
0	
1	

Product Gallery

Brainstorming for data





- Articles by date (latest first)
- Movies by rating / release year
- Songs by artist / title



What type of sort does Spotify use to sort its content?

2. Yes, cycle sort can sort both numeric and non-numeric data

3. Inputting & Displaying Data Complexity

- Inputting data - $O(n)$
- Displaying data - $O(n)$